1 Atmospheric Phenomena and Climate-Model Data

The previous chapters isolate the numerical core of the thesis: randomized approximation, mixed precision, least-squares geometry, and regularized covariance transport. Before returning to the concrete bias-correction workflow, it is useful to make explicit what kind of atmospheric structure the application is meant to represent. The purpose of this chapter is therefore contextual. It does not shift the thesis away from numerical linear algebra; rather, it explains why the covariance and dependence operators used later are meaningful objects in climate-model post-processing.

Atmospheric data are multivariate by nature. Temperature, precipitation, wind, humidity, radiation, pressure, and derived extremes are not independent columns in a table. They are coupled through thermodynamic, radiative, hydrological, and dynamical processes. A statistical correction method that adjusts only marginal distributions may therefore improve each variable separately while leaving the joint state physically implausible. This is the basic reason why multivariate bias correction is a relevant application for the covariance-transport methods studied in this thesis.

1.1 Weather, Climate, and Extremes

The climate-science motivation is strongest for variables and events whose impacts depend on joint behaviour. The IPCC Sixth Assessment Report emphasizes that human-induced climate change is already affecting weather and climate extremes, including hot extremes, heavy precipitation, droughts, storms, and compound events [1]. These phenomena are not only changes in individual means. They involve changes in tails, spatial extent, persistence, co-occurrence, and physical drivers.

Temperature extremes provide the simplest example. A bias correction that matches daily temperature quantiles may still distort the relationship between daily minimum and maximum temperature, humidity, radiation, or wind. Heavy precipitation is similar. Its statistical behaviour depends not only on the marginal distribution of rainfall, but also on circulation, moisture availability, convective activity, orographic forcing, and temporal clustering. Droughts add a further layer: precipitation deficits, evapotranspiration, soil moisture, and radiative forcing interact over longer time scales. Compound events, such as concurrent heat and drought, make the multivariate nature of the problem explicit.

For the present thesis, this motivates a narrow but useful modelling stance. We do not attempt to model all atmospheric dynamics. Instead, we ask whether a dependence-correction layer based on covariance operators can preserve or improve some second-order structure after marginal correction. This is less ambitious than a full physical or distributional model, but it gives a concrete numerical object whose approximation and finite-precision behaviour can be studied.

1.2 Regional Reanalysis and Central Europe

The new local literature also gives a more concrete regional setting. Beranova et al. validate an ALADIN-Climate regional reanalysis over Europe, with emphasis on Central Europe [2]. The model is based on the ALARO configuration, uses a convection-permitting horizontal resolution of 2.3 km, and is evaluated over the period 1990–2014 against ground-based meteorological observations. The variables considered are exactly the kinds of fields that make multivariate post-processing nontrivial: daily temperature, precipitation, relative humidity, wind speed, and global radiation.

The validation results are relevant to the thesis for two reasons. First, they show that high-resolution regional reanalyses can be valuable data sources for Central European climate studies, especially where global reanalyses are too coarse to resolve local terrain and mesoscale structure. Second, they show that even high-resolution reanalyses have systematic, variable-dependent biases. In the ALADIN-Climate validation, mean daily temperature biases are relatively small, but minimum daily temperature is generally overestimated and maximum daily temperature underestimated. Winter precipitation totals are overestimated by nearly 50% on average, partly through an overestimation of wet days. Relative humidity, wind speed, and global radiation also show systematic seasonal and spatial biases [2].

These findings are useful because they give the application chapter a realistic interpretation of the columns in the data matrix. The variables are not abstract features. They are atmospheric quantities with different error mechanisms: microphysics and wet-day frequency for precipitation, boundary-layer and roughness effects for wind, cloud and radiation processes for global radiation, and diurnal-cycle errors for minimum and maximum temperature. A multivariate correction method should therefore be evaluated not only by numerical error, but also by whether it preserves relationships among these physically linked quantities.

1.3 Variables as a Multivariate State

The data model used later can now be read physically. The application chapter uses historical model data, future model data, and historical reference data,

$$X_h \in \mathbb{R}^{n\times p}, \qquad X_f \in \mathbb{R}^{m\times p}, \qquad Y_h \in \mathbb{R}^{n\times p}.$$

The column dimension p is not fixed by the theory. It may encode variables at one site, several nearby locations, lagged time features, seasonal indicators, or blocks of spatial grid points. The same algebraic object therefore covers several application designs:

- intervariable dependence, such as temperature–humidity or precipitation–radiation relationships;
- spatial dependence, such as correlation between nearby grid cells or stations;
- temporal dependence, represented through lagged columns;
- local multivariate states combining variables, locations, and lags.

After marginal correction, the randomized covariance-transport method acts on a latent rank-normal representation. The covariance operators

$$C_X = \frac{1}{n} Z_X^\top Z_X, \qquad C_Y = \frac{1}{n} Z_Y^\top Z_Y$$

should therefore be interpreted as summaries of dependence in this latent state space. They are not a full physical description of the atmosphere. They are a numerically tractable representation of second-order structure after the univariate part of the bias correction has already been handled.

This interpretation also clarifies why the spectrum of the covariance operator matters. Atmospheric fields often contain coherent spatial and intervariable patterns, but they also contain localized extremes and small-scale variability. If the dominant dependence is well approximated by a low-dimensional subspace, then randomized low-rank methods may be effective. If the spectrum is flat or the important structure lies in weak but physically meaningful modes, low-rank compression can fail even when the randomized algorithm behaves as expected mathematically.

1.4 Implications for Covariance Transport

The atmospheric context gives the covariance-transport map a clear but limited role. The map

$$T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$$

is a dependence-correction operator in latent space. It is designed to move the second-order structure of the corrected model sample toward that of the reference sample. It can therefore plausibly affect intervariable correlations, spatial covariance, and some low-order signatures of compound behaviour.

At the same time, its limitations are structural. Covariance transport does not model causality, atmospheric circulation, event sequencing, or nonlinear tail dependence. It does not by itself guarantee that a corrected heatwave, drought, or heavy-precipitation episode is dynamically plausible. This is why the thesis uses covariance transport as a numerical core rather than as a complete climate-science model. The method is valuable only if its narrower target is stated honestly: a compressed and analyzable dependence-correction layer after marginal adjustment.

This also explains the diagnostics used later. Operator-level errors such as $\|C - \widehat{C}\|_2$ and $\|T_\lambda - \widehat{T}_\lambda\|_2$ measure whether the numerical method approximates its own target. Application-facing diagnostics, such as correlation error, spatial or temporal autocorrelation error, and multivariate discrepancy measures, ask whether that target is useful for the atmospheric data. Both levels are necessary. A small transport error is not automatically a successful climate correction, and a useful climate correction may tolerate an operator error that looks large in spectral norm.

1.5 Role in the Thesis

The role of this chapter is to bridge the mathematical and application-facing parts of the thesis. The central contributions remain the perturbation analysis, the randomized low-rank approximation, and the mixed-precision error decomposition. Atmospheric phenomena enter as the reason why the data matrices have multivariate structure and why dependence correction is worth studying.

In the next chapter, this context is specialized to a bias-correction workflow. The atmospheric variables become columns of the data matrices, marginal adjustment is separated from dependence correction, and the covariance-transport operator from ??→p.?? becomes the numerical layer tested in the climate-model application.

References

Back-references to the pages where the publication was cited are given by ●.

[1] IPCC. Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. ed. by Valérie Masson-Delmotte et al. Cambridge University Press, **2021**.
DOI: 10.1017/9781009157896 1

[2] Romana Beranová, Michal Belda, Radmila Brožková, Lucie Pokorná, Jana Popová, Zuzana Rulfová, and Zbyněk Sokol. Evaluating the ALADIN-Climate Model Reanalysis over Central Europe. *Atmospheric Research*, **2025**.
DOI: 10.1016/j.atmosres.2024.107809 2